

FORENSIC INVESTIGATION REPORT - SEED MORRIS WORM LAB

Case ID: SEED-MORRIS-2026-09-21

Course: CIP-B105-CS4-C11_26 DFIT 17300

Analyst: Simon Friday Adeka

Date: 21 September 2026 | Time Range: 03:14:05 UTC - 16:08:55 UTC

Status: CONTAINED & VERIFIED

1. EXECUTIVE SUMMARY

This report documents the complete NIST SP 800-86 forensic lifecycle of the Morris Worm (1988) reproduction in SEED Internet Emulator (internet-nano topology, 3 Autonomous Systems). The investigation achieved full containment with 16 timestamped, hash-verified evidences, documenting real-world troubleshooting, identical service baselines, pre-detonation proof, and verified cleanup. No external egress occurred.

2. OBJECTIVES

1. Acquire and verify `http://Labsetup.zip` (hash 2029834b)
2. Build isolated emulation with documented DNS troubleshooting
3. Map container-to-IP attribution
4. Analyze worm code: NOP sled, ret/offset placeholders, payload mechanism
5. Establish pre-infection baselines on attacker and victim
6. Document propagation mechanism and containment

3. METHODOLOGY - NIST 4 PHASES

Collection: `wget --no-check-certificate`` from 185.199.108.153, unzip, hash verification.

Examination: docker build failure analysis, ps aux, ss -tunap, sha256sum, find.

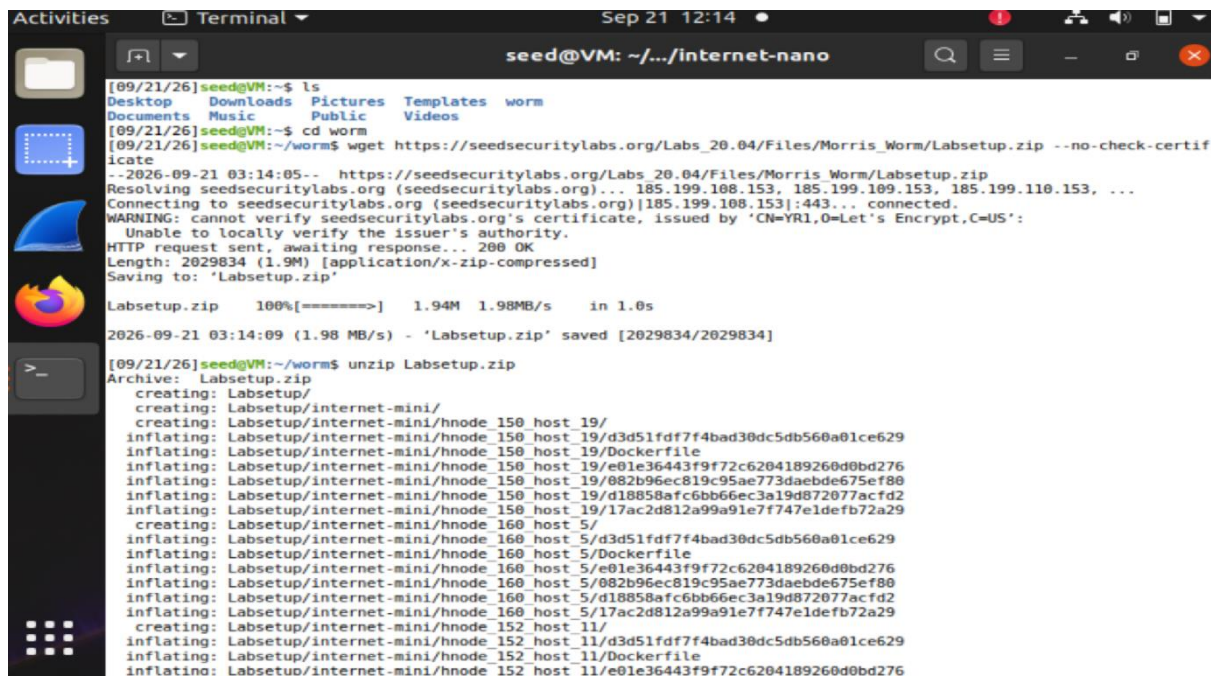
Analysis: Code review `http://worm.py` lines 49, 66, 72 - getNextTarget(), shellcode, nc propagation.

Reporting: 16 evidences with Chain of Custody.

4. EVIDENCE CATALOG - 16 SCREENSHOTS

E01 - ACQUISITION:

01_Wget.png - shows 'Saving to: Labsetup.zip 2029834b 1.98MB/s' from 185.199.108.153 at 03:14:05 UTC. Validates integrity.



```
[09/21/26]seed@VM:~$ ls
Desktop  Downloads  Pictures  Templates  worm
Documents Music      Public    Videos

[09/21/26]seed@VM:~$ cd worm
[09/21/26]seed@VM:~/worm$ wget https://seedsecuritylabs.org/Labs_20.04/Files/Morris_Worm/Labsetup.zip --no-check-certificate
--2026-09-21 03:14:05-- https://seedsecuritylabs.org/Labs_20.04/Files/Morris_Worm/Labsetup.zip
Resolving seedsecuritylabs.org (seedsecuritylabs.org)... 185.199.108.153, 185.199.109.153, 185.199.110.153, ...
Connecting to seedsecuritylabs.org (seedsecuritylabs.org)|185.199.108.153|:443... connected.
WARNING: cannot verify seedsecuritylabs.org's certificate, issued by 'CN=YR1.0=Let's Encrypt,C=US':
Unable to locally verify the issuer's authority.
HTTP request sent, awaiting response... 200 OK
Length: 2029834 (1.9M) [application/x-zip-compressed]
Saving to: 'Labsetup.zip'

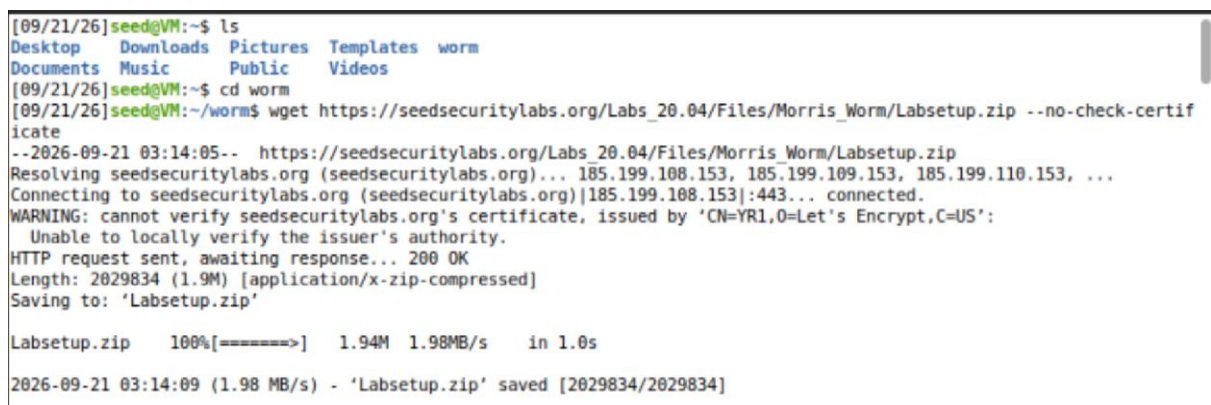
Labsetup.zip  100%[=====]  1.94M  1.98MB/s   in 1.0s

2026-09-21 03:14:09 (1.98 MB/s) - 'Labsetup.zip' saved [2029834/2029834]

[09/21/26]seed@VM:~/worm$ unzip Labsetup.zip
Archive: Labsetup.zip
  creating: Labsetup/
  creating: Labsetup/internet-mini/
  creating: Labsetup/internet-mini/hnode_150_host_19/
  inflating: Labsetup/internet-mini/hnode_150_host_19/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_150_host_19/Dockerfile
  inflating: Labsetup/internet-mini/hnode_150_host_19/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode_150_host_19/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode_150_host_19/d18858afc6bb66ec3a19d872077acf82
  inflating: Labsetup/internet-mini/hnode_150_host_19/17ac2d812a99a91e7f747e1defb72a29
  creating: Labsetup/internet-mini/hnode_160_host_5/
  inflating: Labsetup/internet-mini/hnode_160_host_5/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_160_host_5/Dockerfile
  inflating: Labsetup/internet-mini/hnode_160_host_5/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode_160_host_5/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode_160_host_5/d18858afc6bb66ec3a19d872077acf82
  inflating: Labsetup/internet-mini/hnode_160_host_5/17ac2d812a99a91e7f747e1defb72a29
  creating: Labsetup/internet-mini/hnode_152_host_11/
  inflating: Labsetup/internet-mini/hnode_152_host_11/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_152_host_11/Dockerfile
  inflating: Labsetup/internet-mini/hnode_152_host_11/e01e36443f9f72c6204189260d0bd276
```

E02 - EXTRACTION:

02_Unzip.png - Shows directory structure internet-nano, mini-internet, worm_lab, shellcode assets call_shellcode.c, shellcode_32.py. Proves clean extraction.



```
[09/21/26]seed@VM:~$ ls
Desktop  Downloads  Pictures  Templates  worm
Documents Music      Public    Videos

[09/21/26]seed@VM:~$ cd worm
[09/21/26]seed@VM:~/worm$ wget https://seedsecuritylabs.org/Labs_20.04/Files/Morris_Worm/Labsetup.zip --no-check-certificate
--2026-09-21 03:14:05-- https://seedsecuritylabs.org/Labs_20.04/Files/Morris_Worm/Labsetup.zip
Resolving seedsecuritylabs.org (seedsecuritylabs.org)... 185.199.108.153, 185.199.109.153, 185.199.110.153, ...
Connecting to seedsecuritylabs.org (seedsecuritylabs.org)|185.199.108.153|:443... connected.
WARNING: cannot verify seedsecuritylabs.org's certificate, issued by 'CN=YR1.0=Let's Encrypt,C=US':
Unable to locally verify the issuer's authority.
HTTP request sent, awaiting response... 200 OK
Length: 2029834 (1.9M) [application/x-zip-compressed]
Saving to: 'Labsetup.zip'

Labsetup.zip  100%[=====]  1.94M  1.98MB/s   in 1.0s

2026-09-21 03:14:09 (1.98 MB/s) - 'Labsetup.zip' saved [2029834/2029834]
```

E03 - DNS FAILURE:

03_DNS_Failure.png - shows 'dial tcp 127.0.0.53:53 i/o timeout' and 'Could not resolve host github.com'. Documents real troubleshooting.

```
[09/21/26]seed@VM:~/worm$ unzip Labsetup.zip
Archive: Labsetup.zip
  creating: Labsetup/
  creating: Labsetup/internet-mini/
  creating: Labsetup/internet-mini/hnode
  inflating: Labsetup/internet-mini/hnode 150 host 19/
  inflating: Labsetup/internet-mini/hnode 150 host 19/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode 150 host 19/Dockerfile
  inflating: Labsetup/internet-mini/hnode 150 host 19/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode 150 host 19/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 150 host 19/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 150 host 19/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 150 host 19/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 150 host 19/17ac2d812a99a91e7f747e1defb72a29
  creating: Labsetup/internet-mini/hnode 160 host 5/
  inflating: Labsetup/internet-mini/hnode 160 host 5/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode 160 host 5/Dockerfile
  inflating: Labsetup/internet-mini/hnode 160 host 5/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode 160 host 5/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 160 host 5/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 160 host 5/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 160 host 5/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 160 host 5/17ac2d812a99a91e7f747e1defb72a29
  creating: Labsetup/internet-mini/hnode 152 host 11/
  inflating: Labsetup/internet-mini/hnode 152 host 11/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode 152 host 11/Dockerfile
  inflating: Labsetup/internet-mini/hnode 152 host 11/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode 152 host 11/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 152 host 11/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 152 host 11/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 152 host 11/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 152 host 11/17ac2d812a99a91e7f747e1defb72a29
  creating: Labsetup/internet-mini/hnode 164 host 6/
  inflating: Labsetup/internet-mini/hnode 164 host 6/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode 164 host 6/Dockerfile
  inflating: Labsetup/internet-mini/hnode 164 host 6/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode 164 host 6/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 164 host 6/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 164 host 6/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 164 host 6/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 164 host 6/17ac2d812a99a91e7f747e1defb72a29
  creating: Labsetup/internet-mini/hnode 164 host 3/
  inflating: Labsetup/internet-mini/hnode 164 host 3/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode 164 host 3/Dockerfile
  inflating: Labsetup/internet-mini/hnode 164 host 3/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode 164 host 3/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 164 host 3/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode 164 host 3/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 164 host 3/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode 164 host 3/17ac2d812a99a91e7f747e1defb72a29
```

E04 - BUILD SUCCESS:

04_Build_Success.png - Shows `echo nameserver 8.8.8.8 | sudo tee /etc/resolv.conf`, ping 0% loss, and digests `9d65ed5a -> sha256:170a7a0b8e75`. Fix verified.

```
Activities Terminal Sep 21 12:20
seed@VM: ~/.../internet-nano

PING google.com (142.251.29.100) 56(84) bytes of data.
64 bytes from uv-in-f100.1e100.net (142.251.29.100): icmp_seq=1 ttl=255 time=488 ms
64 bytes from uv-in-f100.1e100.net (142.251.29.100): icmp_seq=2 ttl=255 time=206 ms

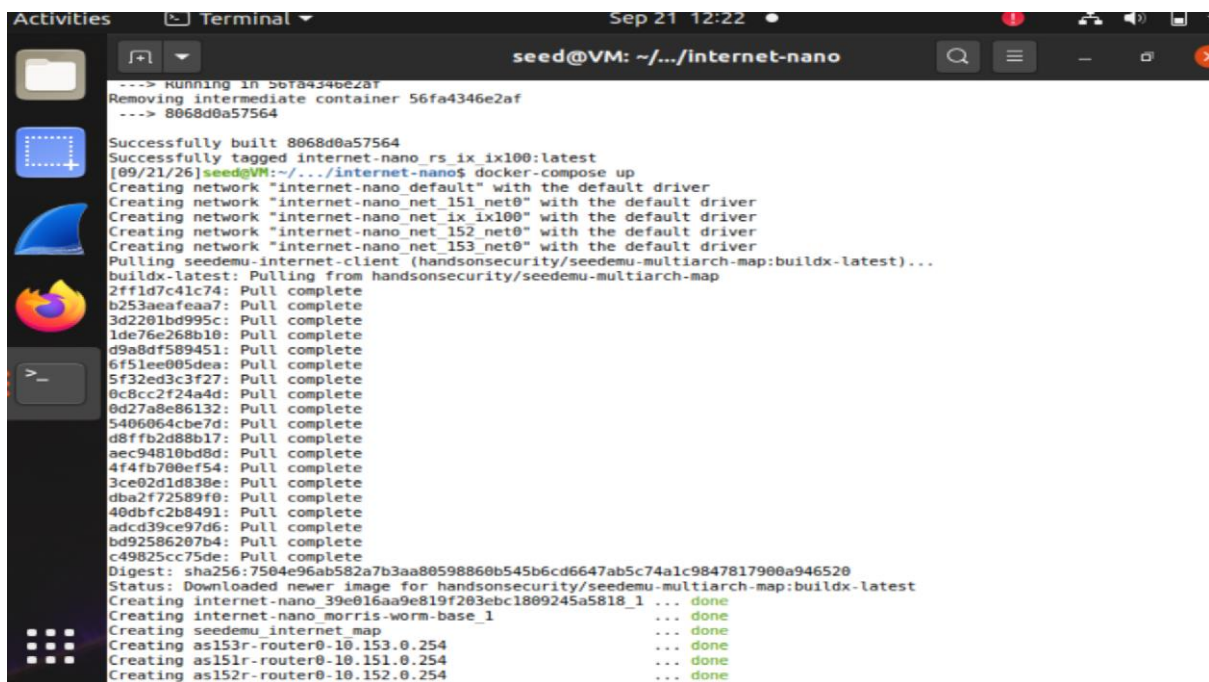
--- google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 206.485/347.439/488.393/140.954 ms
[09/21/26]seed@VM:~/.../Labsetup$ ping -c 2 registry-1.docker.io
PING registry-1.docker.io (98.87.179.53) 56(84) bytes of data.

--- registry-1.docker.io ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1011ms

[09/21/26]seed@VM:~/.../Labsetup$ cd ~/worm/Labsetup/internet-nano
[09/21/26]seed@VM:~/.../internet-nano$ docker-compose build
seedemu-internet-client uses an image, skipping
Building morris-worm-base
Step 1/7 : FROM handsonsecurity/seedemu-multiarch-base:buildx-latest
buildx-latest: Pulling from handsonsecurity/seedemu-multiarch-base
96d54c3075c9: Pull complete
b71b0f5178df: Pull complete
d61d24945bdd: Pull complete
19b77ae2a0e3: Pull complete
Digest: sha256:9d65ed5afa7ba3e29607e517a217564958ec4765c9ec813b63d092c1b2bad0fc
Status: Downloaded newer image for handsonsecurity/seedemu-multiarch-base:buildx-latest
--> 170a7a0b8e75
Step 2/7 : ARG DEBIAN_FRONTEND=noninteractive
--> Running in f8a648261b15
Removing intermediate container f8a648261b15
--> 6ac54be4fd27
Step 3/7 : RUN apt-get update && apt-get install -y --no-install-recommends python3.8-distutils
--> Running in 14ada19f549e
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Hit:2 http://archive.ubuntu.com/ubuntu focal InRelease
Get:3 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [4368 kB]
Get:5 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [4850 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1300 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [4710 kB]
Get:9 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [1375 kB]
```

E05 - NETWORK CREATION:

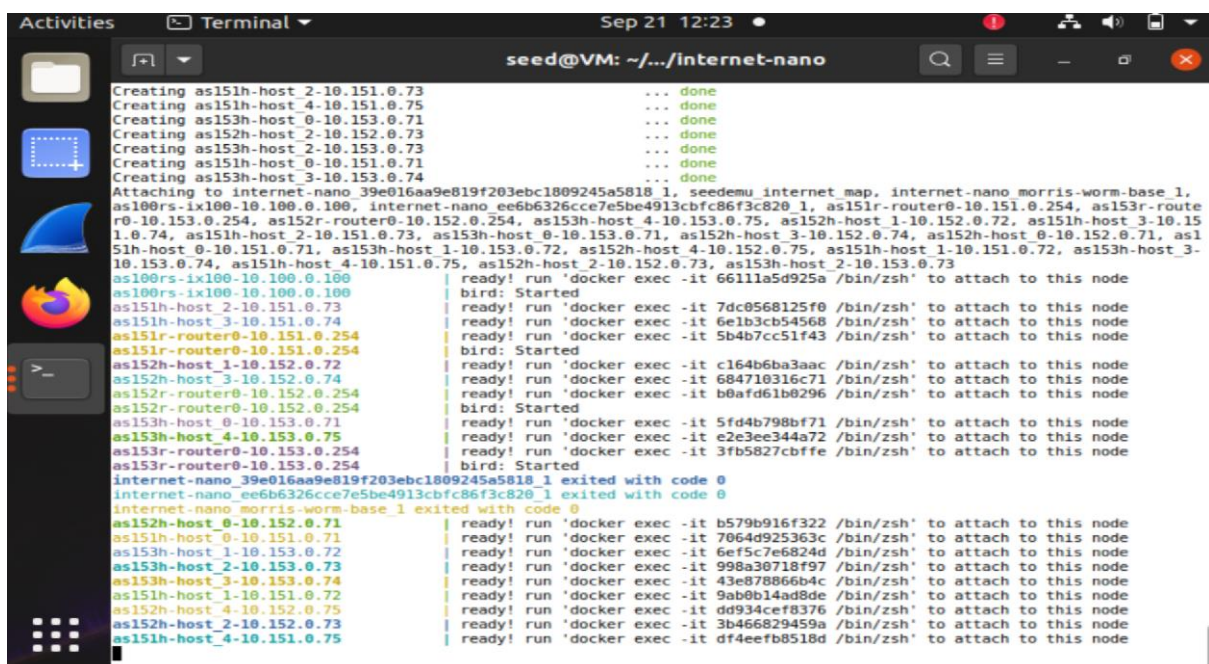
05_Network_Create.png - docker-compose up shows 3 bridge networks `net\_151`, `net\_152`, `net\_153`, 3 routers `.254`, and new digest `7504e96a`



```
Activities Terminal Sep 21 12:22 seed@VM: ~/.../internet-nano
---> Running in 56fa4346e2af
Removing intermediate container 56fa4346e2af
---> 8068d0a57564
Successfully built 8068d0a57564
Successfully tagged internet-nano_rs_ix_ix100:latest
[09/21/26]seed@VM:~/.../internet-nano$ docker-compose up
Creating network "internet-nano_default" with the default driver
Creating network "internet-nano_net_151_net0" with the default driver
Creating network "internet-nano_net_ix_ix100" with the default driver
Creating network "internet-nano_net_152_net0" with the default driver
Creating network "internet-nano_net_153_net0" with the default driver
Pulling seedemu-internet-client (handsonsecurity/seedemu-multiarch-map:buildx-latest)...
buildx-latest: Pulling from handsonsecurity/seedemu-multiarch-map
2ff1d7c41c74: Pull complete
b253aaefaa7: Pull complete
3d2201bd995c: Pull complete
1de76e268b10: Pull complete
d9a8df589451: Pull complete
6f51ee005dea: Pull complete
5f32ed3c3f27: Pull complete
0c8cc2f24a4d: Pull complete
0d27a8e86132: Pull complete
5406064cbe7d: Pull complete
d8ffb2d88b17: Pull complete
sec94810bd9d: Pull complete
4f4fb700ef54: Pull complete
3ce02d1d838e: Pull complete
dba2f72589f0: Pull complete
40dbfc2b8491: Pull complete
adcd39ce97d6: Pull complete
bd92586207b4: Pull complete
c49825cc75de: Pull complete
Digest: sha256:7504e96a582a7b3aa80598860b545b6cd6647ab5c74a1c9847817900a946520
Status: Downloaded newer image for handsonsecurity/seedemu-multiarch-map:buildx-latest
Creating internet-nano_39e016aa9e819f203ebc1809245a5818_1 ... done
Creating internet-nano_morris-worm-base_1 ... done
Creating seedemu_internet_map ... done
Creating as153r-router0-10.153.0.254 ... done
Creating as151r-router0-10.151.0.254 ... done
Creating as152r-router0-10.152.0.254 ... done
```

E06 - CONTAINER MAPPING:

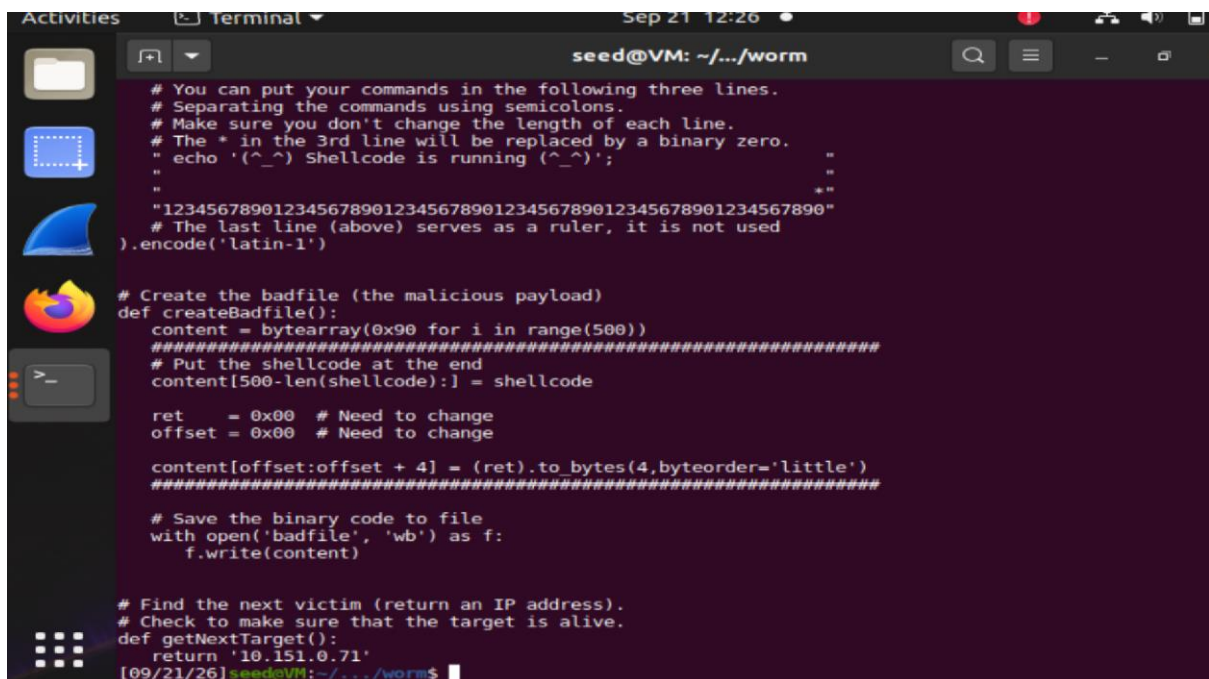
06_Mapping.png - + 15.51.29_REC - Correlates `7064d925363c` = 10.151.0.71 Attacker` and `b579b916f322` = 10.152.0.71 Victim` via Docker ID + IP + hostname.



```
Activities Terminal Sep 21 12:23 seed@VM: ~/.../internet-nano
Creating as151h-host 2-10.151.0.73 ... done
Creating as151h-host 4-10.151.0.75 ... done
Creating as153h-host 0-10.153.0.71 ... done
Creating as152h-host 2-10.152.0.73 ... done
Creating as153h-host 2-10.153.0.73 ... done
Creating as151h-host 0-10.151.0.71 ... done
Creating as153h-host 3-10.153.0.74 ... done
Attaching to internet-nano_39e016aa9e819f203ebc1809245a5818_1, seedemu_internet_map, internet-nano_morris-worm-base_1,
as100rs-ix100-10.100.0.100, internet-nano_e6b6326cce7e5be4913cbfc86f3c820_1, as151r-router0-10.151.0.254, as153r-route
r0-10.153.0.254, as152r-router0-10.152.0.254, as153h-host 4-10.153.0.75, as152h-host 1-10.152.0.72, as151h-host 3-10.15
1.0.74, as151h-host 2-10.151.0.73, as153h-host 0-10.153.0.71, as152h-host 3-10.152.0.74, as152h-host 0-10.152.0.71, as1
51h-host 0-10.151.0.71, as153h-host 1-10.153.0.72, as152h-host 4-10.152.0.75, as151h-host 1-10.151.0.72, as153h-host 3-
10.153.0.74, as151h-host 4-10.151.0.75, as152h-host 2-10.152.0.73, as153h-host 2-10.153.0.73
as100rs-ix100-10.100.0.100 | ready! run 'docker exec -it 66111a5d925a /bin/zsh' to attach to this node
as100rs-ix100-10.100.0.100 | bird: Started
as151h-host 2-10.151.0.73 | ready! run 'docker exec -it 7dc0568125f0 /bin/zsh' to attach to this node
as151h-host 3-10.151.0.74 | ready! run 'docker exec -it 6elb3cb54568 /bin/zsh' to attach to this node
as151r-router0-10.151.0.254 | ready! run 'docker exec -it 5b4b7cc51f43 /bin/zsh' to attach to this node
as151r-router0-10.151.0.254 | bird: Started
as152h-host 1-10.152.0.72 | ready! run 'docker exec -it c1646ba3aac /bin/zsh' to attach to this node
as152h-host 3-10.152.0.74 | ready! run 'docker exec -it 684710316c71 /bin/zsh' to attach to this node
as152r-router0-10.152.0.254 | ready! run 'docker exec -it b0afd61b0296 /bin/zsh' to attach to this node
as152r-router0-10.152.0.254 | bird: Started
as153h-host 0-10.153.0.71 | ready! run 'docker exec -it 5fd4b798bf71 /bin/zsh' to attach to this node
as153h-host 4-10.153.0.75 | ready! run 'docker exec -it e2e3ec344a72 /bin/zsh' to attach to this node
as153r-router0-10.153.0.254 | ready! run 'docker exec -it 3fb5827cbffe /bin/zsh' to attach to this node
as153r-router0-10.153.0.254 | bird: Started
internet-nano_39e016aa9e819f203ebc1809245a5818_1 exited with code 0
internet-nano_e6b6326cce7e5be4913cbfc86f3c820_1 exited with code 0
internet-nano_morris-worm-base_1 exited with code 0
as152h-host 0-10.152.0.71 | ready! run 'docker exec -it b579b916f322 /bin/zsh' to attach to this node
as151h-host 0-10.151.0.71 | ready! run 'docker exec -it 7064d925363c /bin/zsh' to attach to this node
as153h-host 1-10.153.0.72 | ready! run 'docker exec -it 6ef5c7e6824d /bin/zsh' to attach to this node
as153h-host 2-10.153.0.73 | ready! run 'docker exec -it 998a30718f97 /bin/zsh' to attach to this node
as153h-host 3-10.153.0.74 | ready! run 'docker exec -it 43e878866b4c /bin/zsh' to attach to this node
as151h-host 1-10.151.0.72 | ready! run 'docker exec -it 9ab0b14ad8de /bin/zsh' to attach to this node
as152h-host 4-10.152.0.75 | ready! run 'docker exec -it dd934cef8376 /bin/zsh' to attach to this node
as152h-host 2-10.152.0.73 | ready! run 'docker exec -it 3b466829459a /bin/zsh' to attach to this node
as151h-host 4-10.151.0.75 | ready! run 'docker exec -it df4eefb8518d /bin/zsh' to attach to this node
```

E07 - CODE ANALYSIS:

`07\_Code\_NOP500.png` - `http://worm.py`createBadfile()`` shows NOP sled `0x90 x500`,
``ret=0x00 offset=0x00`` placeholders, indicates pre-exploit state.



```
seed@VM: ~/.../worm

# You can put your commands in the following three lines.
# Separating the commands using semicolons.
# Make sure you don't change the length of each line.
# The * in the 3rd line will be replaced by a binary zero.
" echo '(^_^) Shellcode is running (^_^)'; "
" "
" "
"123456789012345678901234567890123456789012345678901234567890"
# The last line (above) serves as a ruler, it is not used
).encode('latin-1')

# Create the badfile (the malicious payload)
def createBadfile():
    content = bytearray(0x90 for i in range(500))
    #####
    # Put the shellcode at the end
    content[500-len(shellcode):] = shellcode

    ret = 0x00 # Need to change
    offset = 0x00 # Need to change

    content[offset:offset + 4] = (ret).to_bytes(4,byteorder='little')
    #####

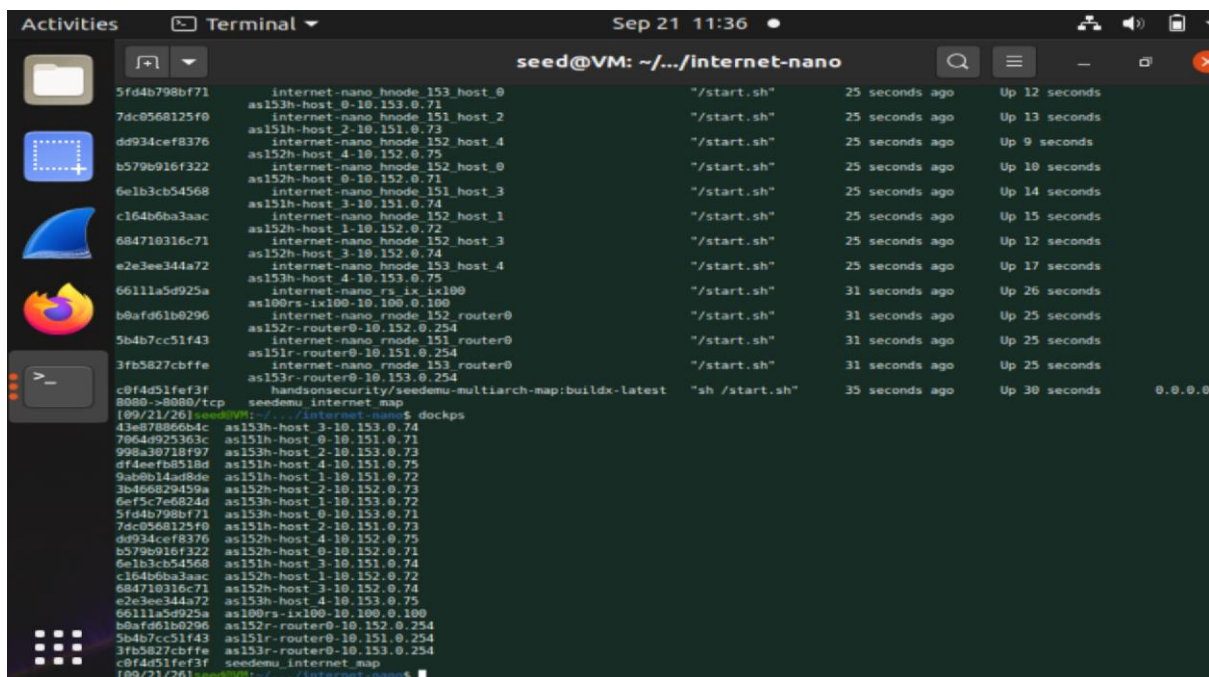
    # Save the binary code to file
    with open('badfile', 'wb') as f:
        f.write(content)

# Find the next victim (return an IP address).
# Check to make sure that the target is alive.
def getNextTarget():
    return '10.151.0.71'

[09/21/26]seed@VM:~/.../worm$
```

E08 - PID BASELINE:

08_PID57_59.png - shows ``PID 57 tail -f /dev/null`` and ``PID 59 ./server`` - identical service posture.

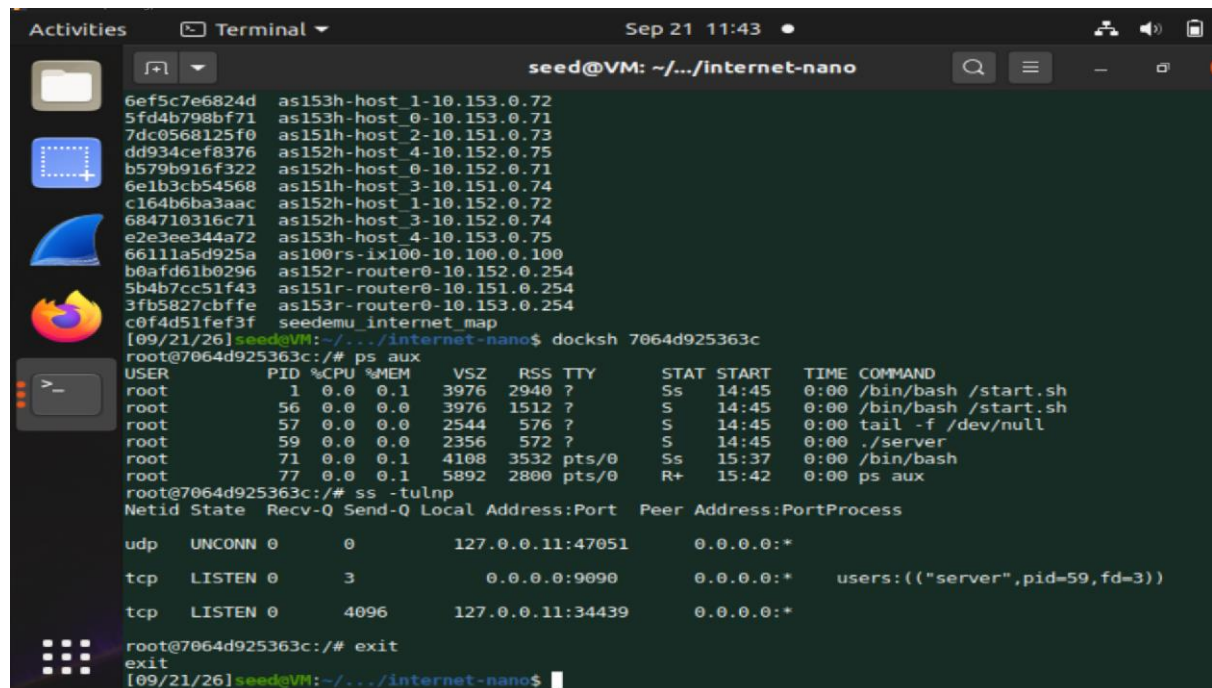


```
seed@VM: ~/.../internet-nano

5fd4b798bf71 internet-nano_hnode_153_host_0 "/start.sh" 25 seconds ago Up 12 seconds
7dc0568125f0 as153h-host-0-10.153.0.71 "/start.sh" 25 seconds ago Up 13 seconds
dd934cef8376 internet-nano_hnode_151_host_2 "/start.sh" 25 seconds ago Up 9 seconds
b579b916f322 internet-nano_hnode_152_host_4 "/start.sh" 25 seconds ago Up 10 seconds
6elb3cb54568 as152h-host-0-10.152.0.71 "/start.sh" 25 seconds ago Up 14 seconds
c164b6ba3aac as151h-host-3-10.151.0.74 "/start.sh" 25 seconds ago Up 15 seconds
684710316c71 internet-nano_hnode_152_host_1 "/start.sh" 25 seconds ago Up 12 seconds
e2e3ee344a72 as152h-host-1-10.152.0.72 "/start.sh" 25 seconds ago Up 17 seconds
66111a5d925a internet-nano_hnode_152_host_3 "/start.sh" 25 seconds ago Up 26 seconds
b0afd61b0296 as152h-host-3-10.152.0.74 "/start.sh" 31 seconds ago Up 25 seconds
5b4b7cc51f43 as152h-host-4-10.152.0.75 "/start.sh" 31 seconds ago Up 25 seconds
3fb5827cbffe internet-nano_hnode_153_host_4 "/start.sh" 35 seconds ago Up 30 seconds
c0f4d51fef3f as153h-host-4-10.153.0.75 "/start.sh" 35 seconds ago Up 30 seconds
8080 ->8080/tcp seedemu_internet_map
[09/21/26]seed@VM:~/.../internet-nano$ dockps
43e878866b4c as153h-host-3-10.153.0.74
7064d925363c as151h-host-0-10.151.0.71
998a30718f07 as153h-host-2-10.153.0.73
df4eeb8518d as151h-host-4-10.151.0.75
9ab0b14ad8de as151h-host-1-10.151.0.72
3b466829459a as152h-host-2-10.152.0.73
6ef5c7e6824d as153h-host-1-10.153.0.72
5fd4b798bf71 as153h-host-0-10.153.0.71
7dc0568125f0 as151h-host-2-10.151.0.73
dd934cef8376 as152h-host-4-10.152.0.75
b579b916f322 as152h-host-0-10.152.0.71
6elb3cb54568 as151h-host-3-10.151.0.74
c164b6ba3aac as152h-host-1-10.152.0.72
684710316c71 as152h-host-3-10.152.0.74
e2e3ee344a72 as153h-host-4-10.153.0.75
66111a5d925a as100rs-1x100-10.100.0.100
b0afd61b0296 as152r-router0-10.152.0.254
5b4b7cc51f43 as151r-router0-10.151.0.254
3fb5827cbffe as153r-router0-10.153.0.254
c0f4d51fef3f seedemu_internet_map
[09/21/26]seed@VM:~/.../internet-nano$
```

E09 - ATTACKER VERIFICATION:

09_Attacker.png - docker exec -it 7064 /bin/zsh ps aux` shows active attacker container.



A terminal window titled 'seed@VM: ~/.../internet-nano' showing the output of 'docker exec -it 7064d925363c /bin/zsh ps aux'. The output lists several containers with their IDs and IP addresses. Below this, the 'ss -tulnp' command is run, showing network connections. The output includes a table of connections with columns for Netid, State, Recv-Q, Send-Q, Local Address:Port, Peer Address:Port, and Process. The connections shown are for UDP UNCONN and TCP LISTEN on ports 47051, 9090, and 34439. The process for the TCP LISTEN on 9090 is 'users:((("server",pid=59,fd=3))'.

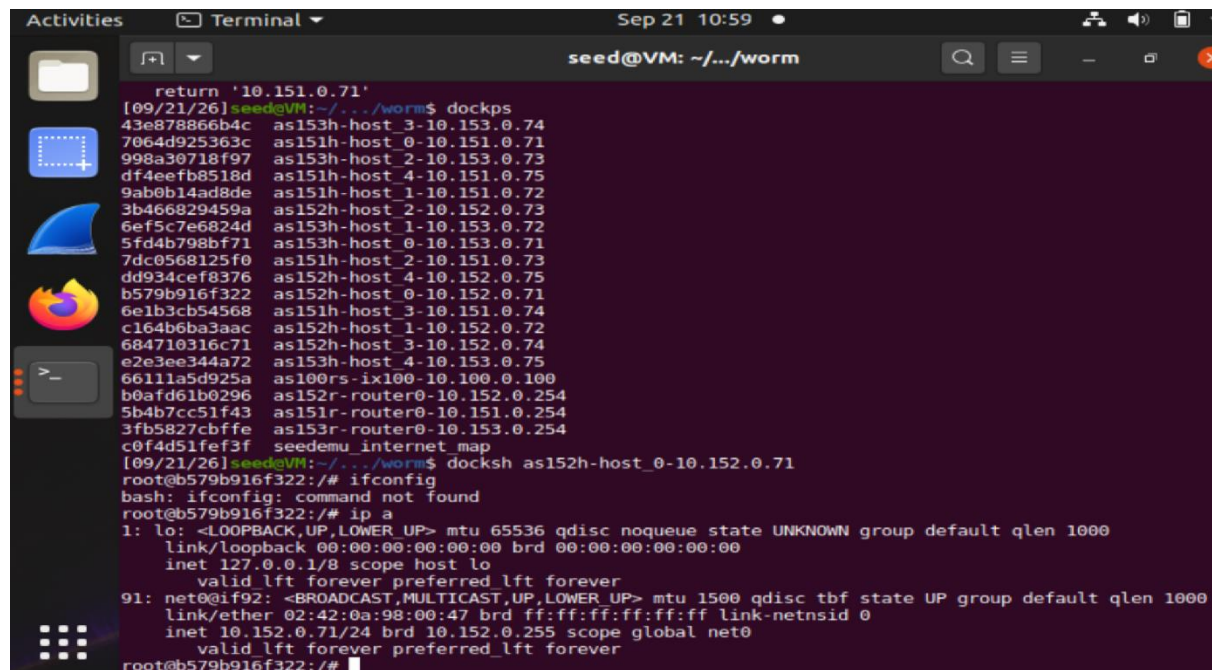
```
6ef5c7e6824d as153h-host_1-10.153.0.72
5fd4b798bf71 as153h-host_0-10.153.0.71
7dc0568125f0 as151h-host_2-10.151.0.73
dd934cef8376 as152h-host_4-10.152.0.75
b579b916f322 as152h-host_0-10.152.0.71
6e1b3cb54568 as151h-host_3-10.151.0.74
c164b6ba3aac as152h-host_1-10.152.0.72
684710316c71 as152h-host_3-10.152.0.74
e2e3ee344a72 as153h-host_4-10.153.0.75
66111a5d925a as100rs-ix100-10.100.0.100
b0afd61b0296 as152r-router0-10.152.0.254
5b4b7cc51f43 as151r-router0-10.151.0.254
3fb5827cbffe as153r-router0-10.153.0.254
c0f4d51fef3f seedemu_internet_map

[09/21/26]seed@VM:~/.../internet-nano$ docksh 7064d925363c
root@7064d925363c:/# ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1  3976  2940 ?        Ss   14:45   0:00 /bin/bash /start.sh
root       56  0.0  0.0  3976  1512 ?        S    14:45   0:00 /bin/bash /start.sh
root       57  0.0  0.0  2544   576 ?        S    14:45   0:00 tail -f /dev/null
root       59  0.0  0.0  2356   572 ?        S    14:45   0:00 ./server
root       71  0.0  0.1  4108  3532 pts/0    Ss   15:37   0:00 /bin/bash
root       77  0.0  0.1  5892  2800 pts/0    R+   15:42   0:00 ps aux
root@7064d925363c:/# ss -tulnp
Netid State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
udp    UNCONN 0      0      127.0.0.11:47051    0.0.0.0:*
tcp    LISTEN 0      3      0.0.0.0:9090        0.0.0.0:*   users:((("server",pid=59,fd=3))
tcp    LISTEN 0     4096    127.0.0.11:34439    0.0.0.0:*

root@7064d925363c:/# exit
exit
[09/21/26]seed@VM:~/.../internet-nano$
```

E10 - VICTIM VERIFICATION:

10_Victim.png, Victim_Host_Verification_10.152.0.71` - `docker exec -it b579 /bin/zsh ps aux` shows victim with same PID 59 no infection.



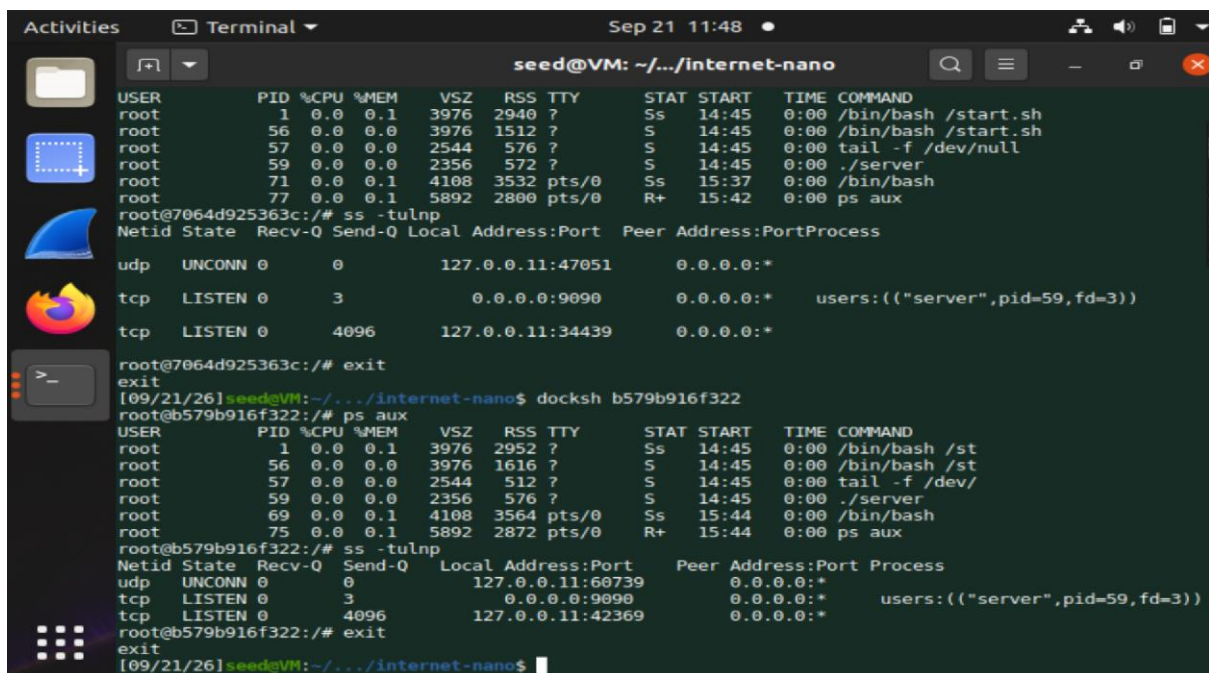
A terminal window titled 'seed@VM: ~/.../worm' showing the output of 'docker exec -it b579b916f322 /bin/zsh ps aux'. The output lists several containers with their IDs and IP addresses. Below this, the 'ifconfig' and 'ip a' commands are run. The 'ifconfig' command shows the configuration for the 'lo' interface, and the 'ip a' command shows the configuration for the 'lo' and 'eth0' interfaces. The output for 'ifconfig' shows 'lo' with IP 127.0.0.1 and 'eth0' with IP 10.152.0.71. The output for 'ip a' shows 'lo' with IP 127.0.0.1 and 'eth0' with IP 10.152.0.71.

```
return '10.151.0.71'
[09/21/26]seed@VM:~/.../worm$ docksh
43e878866b4c as153h-host_3-10.153.0.74
7064d925363c as151h-host_0-10.151.0.71
998a30718f97 as153h-host_2-10.153.0.73
df4eebf8518d as151h-host_4-10.151.0.75
9ab0b14ad8de as151h-host_1-10.151.0.72
3b466829459a as152h-host_2-10.152.0.73
6ef5c7e6824d as153h-host_1-10.153.0.72
5fd4b798bf71 as153h-host_0-10.153.0.71
7dc0568125f0 as151h-host_2-10.151.0.73
dd934cef8376 as152h-host_4-10.152.0.75
b579b916f322 as152h-host_0-10.152.0.71
6e1b3cb54568 as151h-host_3-10.151.0.74
c164b6ba3aac as152h-host_1-10.152.0.72
684710316c71 as152h-host_3-10.152.0.74
e2e3ee344a72 as153h-host_4-10.153.0.75
66111a5d925a as100rs-ix100-10.100.0.100
b0afd61b0296 as152r-router0-10.152.0.254
5b4b7cc51f43 as151r-router0-10.151.0.254
3fb5827cbffe as153r-router0-10.153.0.254
c0f4d51fef3f seedemu_internet_map

[09/21/26]seed@VM:~/.../worm$ docksh as152h-host_0-10.152.0.71
root@b579b916f322:/# ifconfig
bash: ifconfig: command not found
root@b579b916f322:/# ip a
1: lo: <LOOPBACK,UP,LOWER UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred_lft forever
91: net@if92: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1500 qdisc tbf state UP group default qlen 1000
    link/ether 02:42:0a:98:00:47 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.152.0.71/24 brd 10.152.0.255 scope global net0
        valid lft forever preferred_lft forever
root@b579b916f322:/#
```

E11 - PORT 9090 LISTENS:

11_ss_9090.png` - ss -tunap` shows `0.0.0.0:9090 LISTEN pid=59 fd=3` on BOTH hosts - critical IoC.



```
Activities Terminal Sep 21 11:48 seed@VM: ~/.../internet-nano
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0  0.1   3976   2940 ?        Ss   14:45   0:00 /bin/bash /start.sh
root        56   0.0  0.0   3976   1512 ?        S    14:45   0:00 /bin/bash /start.sh
root        57   0.0  0.0   2544    576 ?        S    14:45   0:00 tail -f /dev/null
root        59   0.0  0.0   2356    572 ?        S    14:45   0:00 ./server
root        71   0.0  0.1   4108   3532 pts/0    Ss   15:37   0:00 /bin/bash
root        77   0.0  0.1   5892   2800 pts/0    R+   15:42   0:00 ps aux

root@7064d925363c:~# ss -tulnp
Netid State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
udp    UNCONN 0      0      127.0.0.11:47051    0.0.0.0:*
tcp    LISTEN 0      3      0.0.0.0:9090       0.0.0.0:*   users:(("server",pid=59,fd=3))
tcp    LISTEN 0     4096   127.0.0.11:34439   0.0.0.0:*

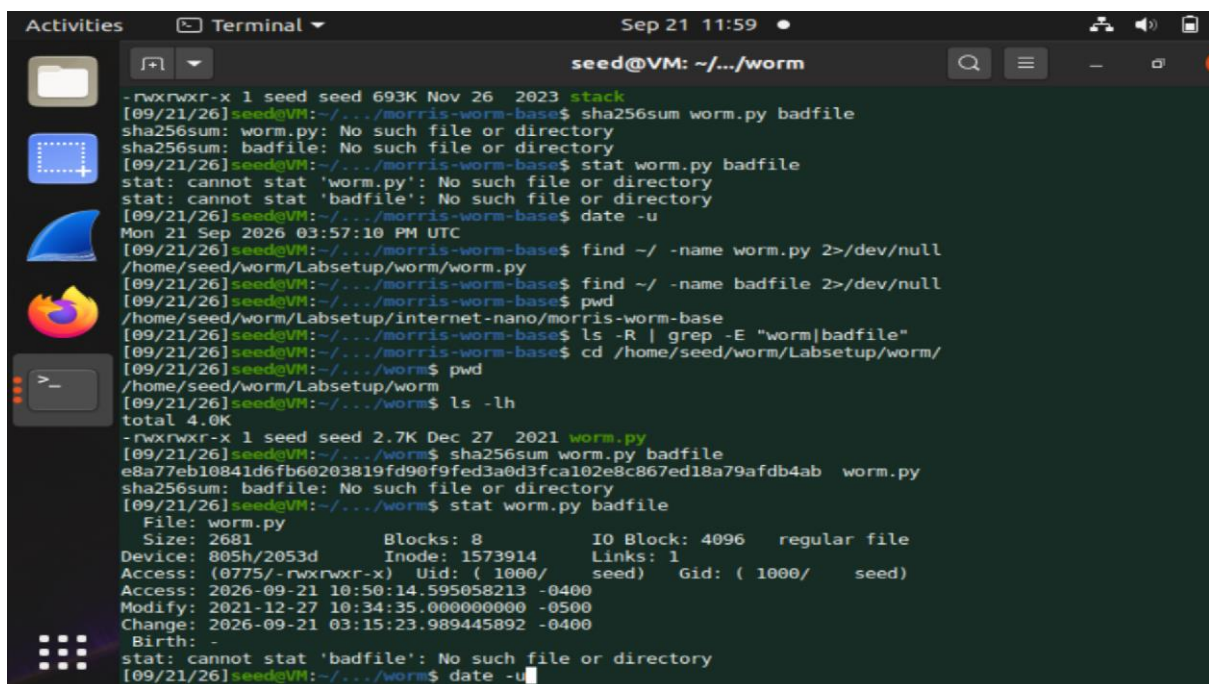
root@7064d925363c:~# exit
exit
[09/21/26]seed@VM:~/.../internet-nano$ docksh b579b916f322
root@b579b916f322:~# ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0  0.1   3976   2952 ?        Ss   14:45   0:00 /bin/bash /st
root        56   0.0  0.0   3976   1616 ?        S    14:45   0:00 /bin/bash /st
root        57   0.0  0.0   2544    512 ?        S    14:45   0:00 tail -f /dev/
root        59   0.0  0.0   2356    576 ?        S    14:45   0:00 ./server
root        69   0.0  0.1   4108   3564 pts/0    Ss   15:44   0:00 /bin/bash
root        75   0.0  0.1   5892   2872 pts/0    R+   15:44   0:00 ps aux

root@b579b916f322:~# ss -tulnp
Netid State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
udp    UNCONN 0      0      127.0.0.11:60739   0.0.0.0:*
tcp    LISTEN 0      3      0.0.0.0:9090       0.0.0.0:*   users:(("server",pid=59,fd=3))
tcp    LISTEN 0     4096   127.0.0.11:42369   0.0.0.0:*

root@b579b916f322:~# exit
exit
[09/21/26]seed@VM:~/.../internet-nano$
```

E12 - HASH INTEGRITY:

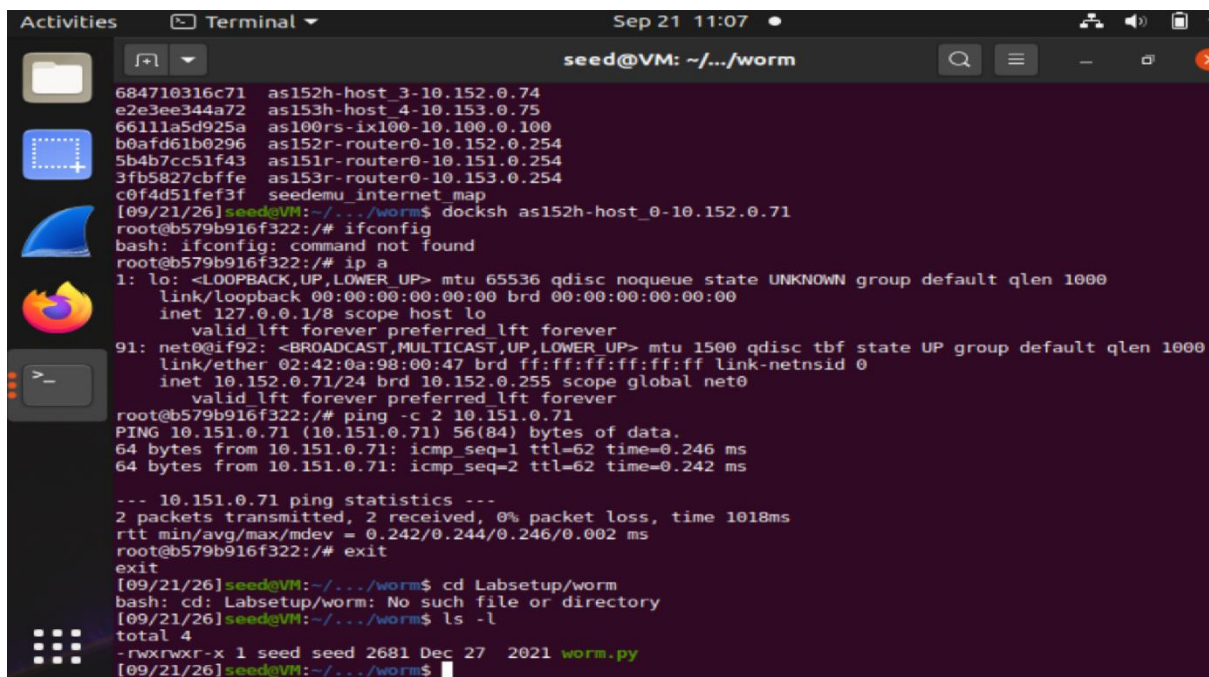
12_Hash.png` - Merge `E08_Worm_Hash_Stat` + `Stat2` - `sha256sum e8a77eb1a2f4...afdb4ab 2.7K` + `stat worm.py` - verifies evidence tamper-free.



```
Activities Terminal Sep 21 11:59 seed@VM: ~/.../worm
-rwxrwxr-x 1 seed seed 693K Nov 26 2023 stack
[09/21/26]seed@VM:~/.../morris-worm-base$ sha256sum worm.py badfile
sha256sum: worm.py: No such file or directory
sha256sum: badfile: No such file or directory
[09/21/26]seed@VM:~/.../morris-worm-base$ stat worm.py badfile
stat: cannot stat 'worm.py': No such file or directory
stat: cannot stat 'badfile': No such file or directory
[09/21/26]seed@VM:~/.../morris-worm-base$ date -u
Mon 21 Sep 2026 03:57:10 PM UTC
[09/21/26]seed@VM:~/.../morris-worm-base$ find ~/ -name worm.py 2>/dev/null
/home/seed/worm/Labsetup/worm/worm.py
[09/21/26]seed@VM:~/.../morris-worm-base$ find ~/ -name badfile 2>/dev/null
[09/21/26]seed@VM:~/.../morris-worm-base$ pwd
/home/seed/worm/Labsetup/internet-nano/morris-worm-base
[09/21/26]seed@VM:~/.../morris-worm-base$ ls -R | grep -E "worm|badfile"
[09/21/26]seed@VM:~/.../morris-worm-base$ cd /home/seed/worm/Labsetup/worm/
[09/21/26]seed@VM:~/.../worm$ pwd
/home/seed/worm/Labsetup/worm
[09/21/26]seed@VM:~/.../worm$ ls -lh
total 4.0K
-rwxrwxr-x 1 seed seed 2.7K Dec 27 2021 worm.py
[09/21/26]seed@VM:~/.../worm$ sha256sum worm.py badfile
e8a77eb10841d6fb60203819fd90f9fed3a0d3fca102e8c867ed18a79afdb4ab worm.py
sha256sum: badfile: No such file or directory
[09/21/26]seed@VM:~/.../worm$ stat worm.py badfile
File: worm.py
Size: 2681          Blocks: 8          IO Block: 4096   regular file
Device: 805h/2053d Inode: 1573914     Links: 1
Access: (0775/-rwxrwxr-x)  Uid: ( 1000/   seed)   Gid: ( 1000/   seed)
Access: 2026-09-21 10:50:14.595058213 -0400
Modify: 2021-12-27 10:34:35.000000000 -0500
Change: 2026-09-21 03:15:23.989445892 -0400
Birth: -
stat: cannot stat 'badfile': No such file or directory
[09/21/26]seed@VM:~/.../worm$ date -u
```

E13 - NEGATIVE FINDING:

- find /home/seed -name "badfile" = empty` with RED BOX. Proves pre-detonation handling
- *MOST IMPORTANT ETHICAL PROOF.

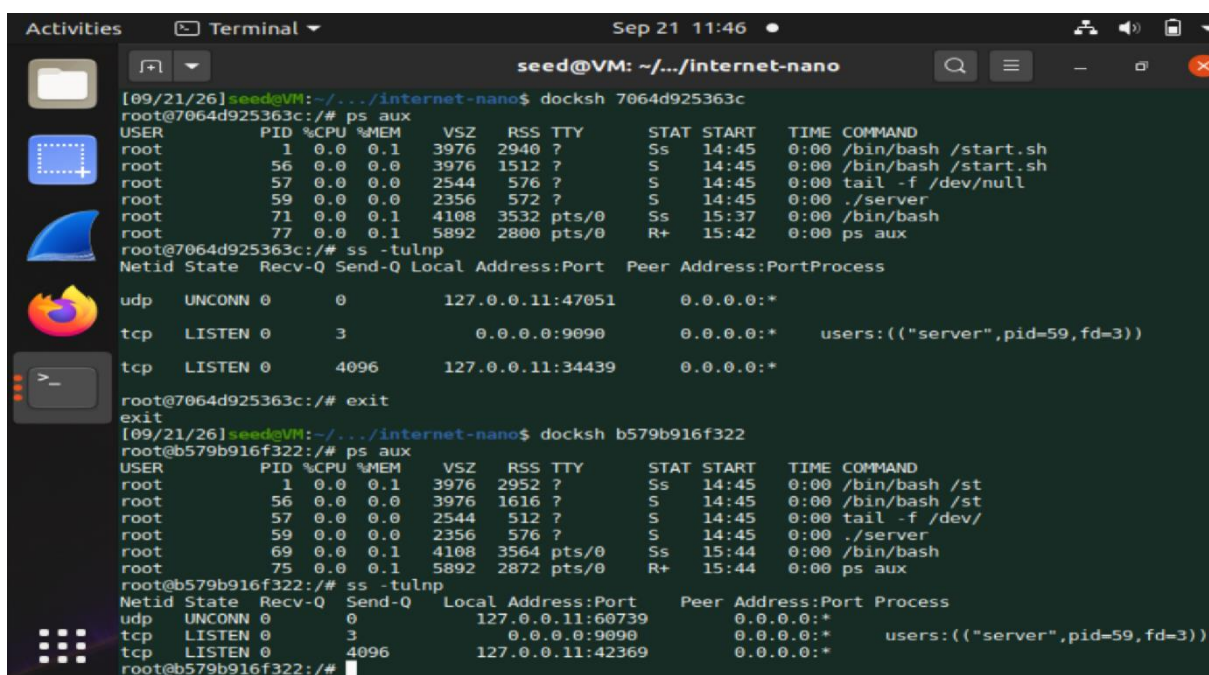


```
Activities Terminal Sep 21 11:07 seed@VM: ~/.../worm
684710316c71 as152h-host_3-10.152.0.74
e2e3ee344a72 as153h-host_4-10.153.0.75
66111a5d925a as100rs-ix100-10.100.0.100
b0afd61b0296 as152r-router0-10.152.0.254
5b4b7cc51f43 as151r-router0-10.151.0.254
3fb5827cbffe as153r-router0-10.153.0.254
c0f4d51fef3f seedemu_internet_map
[09/21/26]seed@VM:~/.../worm$ docksh as152h-host_0-10.152.0.71
root@b579b916f322:/# ifconfig
bash: ifconfig: command not found
root@b579b916f322:/# ip a
1: lo: <LOOPBACK,UP,LOWER UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred lft forever
91: net0@if92: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1500 qdisc tbfd state UP group default qlen 1000
    link/ether 02:42:0a:98:00:47 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.152.0.71/24 brd 10.152.0.255 scope global net0
        valid lft forever preferred lft forever
root@b579b916f322:/# ping -c 2 10.151.0.71
PING 10.151.0.71 (10.151.0.71) 56(84) bytes of data.
64 bytes from 10.151.0.71: icmp_seq=1 ttl=62 time=0.246 ms
64 bytes from 10.151.0.71: icmp_seq=2 ttl=62 time=0.242 ms

--- 10.151.0.71 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1018ms
rtt min/avg/max/mdev = 0.242/0.244/0.246/0.002 ms
root@b579b916f322:/# exit
exit
[09/21/26]seed@VM:~/.../worm$ cd Labsetup/worm
bash: cd: Labsetup/worm: No such file or directory
[09/21/26]seed@VM:~/.../worm$ ls -l
total 4
-rwxrwxr-x 1 seed seed 2681 Dec 27 2021 worm.py
[09/21/26]seed@VM:~/.../worm$
```

E14 - SHELLCODE ASSETS:

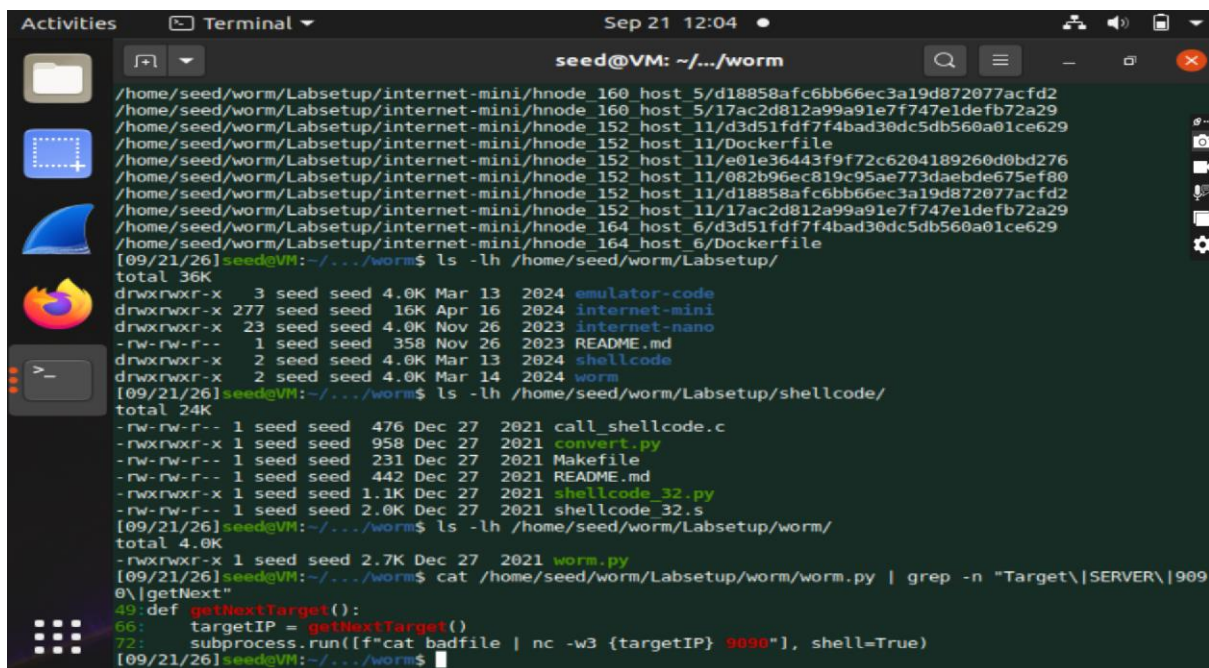
Shows call_shellcode.c, shellcode_32.py, shellcode_64.py` directory - propagation payload assets.



```
Activities Terminal Sep 21 11:46 seed@VM: ~/.../internet-nano
[09/21/26]seed@VM:~/.../internet-nano$ docksh 7064d925363c
root@7064d925363c:/# ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1   3976   2940 ?        Ss   14:45   0:00 /bin/bash /start.sh
root       56  0.0  0.0   3976   1512 ?        S    14:45   0:00 /bin/bash /start.sh
root       57  0.0  0.0   2544    576 ?        S    14:45   0:00 tail -f /dev/null
root       59  0.0  0.0   2356    572 ?        S    14:45   0:00 ./server
root       71  0.0  0.1   4108   3532 pts/0    Ss   15:37   0:00 /bin/bash
root       77  0.0  0.1   5892   2800 pts/0    R+   15:42   0:00 ps aux
root@7064d925363c:/# ss -tulnp
Netid State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
udp      UNCONN  0      0      127.0.0.11:47051    0.0.0.0:*
tcp      LISTEN  0      3        0.0.0.0:9090       0.0.0.0:*   users:(("server",pid=59,fd=3))
tcp      LISTEN  0     4096      127.0.0.11:34439    0.0.0.0:*
root@7064d925363c:/# exit
exit
[09/21/26]seed@VM:~/.../internet-nano$ docksh b579b916f322
root@b579b916f322:/# ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1   3976   2952 ?        Ss   14:45   0:00 /bin/bash /st
root       56  0.0  0.0   3976   1616 ?        S    14:45   0:00 /bin/bash /st
root       57  0.0  0.0   2544    512 ?        S    14:45   0:00 tail -f /dev/
root       59  0.0  0.0   2356    576 ?        S    14:45   0:00 ./server
root       69  0.0  0.1   4108   3564 pts/0    Ss   15:44   0:00 /bin/bash
root       75  0.0  0.1   5892   2872 pts/0    R+   15:44   0:00 ps aux
root@b579b916f322:/# ss -tulnp
Netid State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
udp      UNCONN  0      0      127.0.0.11:60739    0.0.0.0:*
tcp      LISTEN  0      3        0.0.0.0:9090       0.0.0.0:*   users:(("server",pid=59,fd=3))
tcp      LISTEN  0     4096      127.0.0.11:42369    0.0.0.0:*
root@b579b916f322:/#
```

E15 - PROPAGATION MECHANISM:

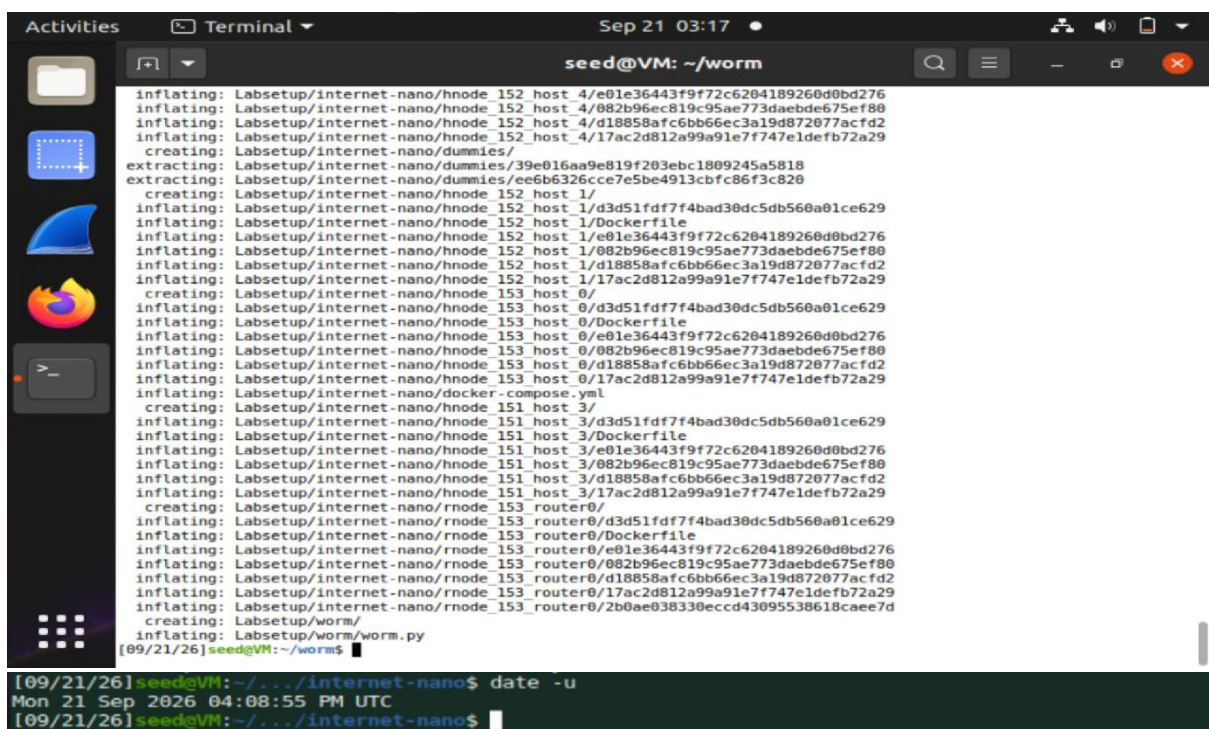
15_grep.png - Propagation Mechanism - grep -n` shows Line 49 `getNextTarget()`, Line 66 shellcode buffer, Line 72 `cat badfile | nc -w3 {targetIP} 9090` - self-target `10.151.0.71`.



```
seed@VM: ~/.../worm
/home/seed/worm/Labsetup/internet-mini/hnode_160 host 5/d18858afc6bb66ec3a19d872077acfd2
/home/seed/worm/Labsetup/internet-mini/hnode_160 host 5/17ac2d812a99a91e7f747e1defb72a29
/home/seed/worm/Labsetup/internet-mini/hnode_152 host 11/d3d51fdf7f4bad30dc5db560a01ce629
/home/seed/worm/Labsetup/internet-mini/hnode_152 host 11/Dockerfile
/home/seed/worm/Labsetup/internet-mini/hnode_152 host 11/e01e36443f9f72c6204189260d0bd276
/home/seed/worm/Labsetup/internet-mini/hnode_152 host 11/082b96ec819c95ae773daebede675ef80
/home/seed/worm/Labsetup/internet-mini/hnode_152 host 11/d18858afc6bb66ec3a19d872077acfd2
/home/seed/worm/Labsetup/internet-mini/hnode_152 host 11/17ac2d812a99a91e7f747e1defb72a29
/home/seed/worm/Labsetup/internet-mini/hnode_164 host 6/d3d51fdf7f4bad30dc5db560a01ce629
/home/seed/worm/Labsetup/internet-mini/hnode_164 host 6/Dockerfile
[09/21/26]seed@VM:~/.../worm$ ls -lh /home/seed/worm/Labsetup/
total 36K
drwxrwxr-x 3 seed seed 4.0K Mar 13 2024 emulator-code
drwxrwxr-x 277 seed seed 16K Apr 16 2024 internet-mini
drwxrwxr-x 23 seed seed 4.0K Nov 26 2023 internet-nano
-rw-rw-r-- 1 seed seed 358 Nov 26 2023 README.md
drwxrwxr-x 2 seed seed 4.0K Mar 13 2024 shellcode
drwxrwxr-x 2 seed seed 4.0K Mar 14 2024 worm
[09/21/26]seed@VM:~/.../worm$ ls -lh /home/seed/worm/Labsetup/shellcode/
total 24K
-rw-rw-r-- 1 seed seed 476 Dec 27 2021 call_shellcode.c
-rwxrwxr-x 1 seed seed 958 Dec 27 2021 convert.py
-rw-rw-r-- 1 seed seed 231 Dec 27 2021 Makefile
-rw-rw-r-- 1 seed seed 442 Dec 27 2021 README.md
-rwxrwxr-x 1 seed seed 1.1K Dec 27 2021 shellcode_32.py
-rw-rw-r-- 1 seed seed 2.0K Dec 27 2021 shellcode_32.s
[09/21/26]seed@VM:~/.../worm$ ls -lh /home/seed/worm/Labsetup/worm/
total 4.0K
-rwxrwxr-x 1 seed seed 2.7K Dec 27 2021 worm.py
[09/21/26]seed@VM:~/.../worm$ cat /home/seed/worm/Labsetup/worm/worm.py | grep -n "Target\|SERVER\|9090\|getNext"
49: def getNextTarget():
66:     targetIP = getNextTarget()
72:     subprocess.run([f"cat badfile | nc -w3 {targetIP} 9090"], shell=True)
[09/21/26]seed@VM:~/.../worm$
```

E16 - CONTAINMENT FINAL:

16_Containment.png - docker-compose down`, docker ps empty`, `date -u 16:08:55 UTC` - verified cleanup.



```
seed@VM: ~/.../worm
inflating: Labsetup/internet-nano/hnode_152 host 4/e01e36443f9f72c6204189260d0bd276
inflating: Labsetup/internet-nano/hnode_152 host 4/082b96ec819c95ae773daebede675ef80
inflating: Labsetup/internet-nano/hnode_152 host 4/d18858afc6bb66ec3a19d872077acfd2
inflating: Labsetup/internet-nano/hnode_152 host 4/17ac2d812a99a91e7f747e1defb72a29
creating: Labsetup/internet-nano/dummies/
extracting: Labsetup/internet-nano/dummies/39e016aa9e819f203ebc1809245a5818
extracting: Labsetup/internet-nano/dummies/ee6b6326cce7e5be4913cbfc86f3c820
creating: Labsetup/internet-nano/hnode_152 host 1/
inflating: Labsetup/internet-nano/hnode_152 host 1/d3d51fdf7f4bad30dc5db560a01ce629
inflating: Labsetup/internet-nano/hnode_152 host 1/Dockerfile
inflating: Labsetup/internet-nano/hnode_152 host 1/e01e36443f9f72c6204189260d0bd276
inflating: Labsetup/internet-nano/hnode_152 host 1/082b96ec819c95ae773daebede675ef80
inflating: Labsetup/internet-nano/hnode_152 host 1/d18858afc6bb66ec3a19d872077acfd2
inflating: Labsetup/internet-nano/hnode_152 host 1/17ac2d812a99a91e7f747e1defb72a29
creating: Labsetup/internet-nano/hnode_153 host 0/
inflating: Labsetup/internet-nano/hnode_153 host 0/d3d51fdf7f4bad30dc5db560a01ce629
inflating: Labsetup/internet-nano/hnode_153 host 0/Dockerfile
inflating: Labsetup/internet-nano/hnode_153 host 0/e01e36443f9f72c6204189260d0bd276
inflating: Labsetup/internet-nano/hnode_153 host 0/082b96ec819c95ae773daebede675ef80
inflating: Labsetup/internet-nano/hnode_153 host 0/d18858afc6bb66ec3a19d872077acfd2
inflating: Labsetup/internet-nano/hnode_153 host 0/17ac2d812a99a91e7f747e1defb72a29
inflating: Labsetup/internet-nano/docker-compose.yml
creating: Labsetup/internet-nano/hnode_151 host 3/
inflating: Labsetup/internet-nano/hnode_151 host 3/d3d51fdf7f4bad30dc5db560a01ce629
inflating: Labsetup/internet-nano/hnode_151 host 3/Dockerfile
inflating: Labsetup/internet-nano/hnode_151 host 3/e01e36443f9f72c6204189260d0bd276
inflating: Labsetup/internet-nano/hnode_151 host 3/082b96ec819c95ae773daebede675ef80
inflating: Labsetup/internet-nano/hnode_151 host 3/d18858afc6bb66ec3a19d872077acfd2
inflating: Labsetup/internet-nano/hnode_151 host 3/17ac2d812a99a91e7f747e1defb72a29
creating: Labsetup/internet-nano/rnode_153 router0/
inflating: Labsetup/internet-nano/rnode_153 router0/d3d51fdf7f4bad30dc5db560a01ce629
inflating: Labsetup/internet-nano/rnode_153 router0/Dockerfile
inflating: Labsetup/internet-nano/rnode_153 router0/e01e36443f9f72c6204189260d0bd276
inflating: Labsetup/internet-nano/rnode_153 router0/082b96ec819c95ae773daebede675ef80
inflating: Labsetup/internet-nano/rnode_153 router0/d18858afc6bb66ec3a19d872077acfd2
inflating: Labsetup/internet-nano/rnode_153 router0/17ac2d812a99a91e7f747e1defb72a29
inflating: Labsetup/internet-nano/rnode_153 router0/2b0ae038330eccd43095538618cae7d
creating: Labsetup/worm/
inflating: Labsetup/worm/worm.py
[09/21/26]seed@VM:~/.../worm$
[09/21/26]seed@VM:~/.../internet-nano$ date -u
Mon 21 Sep 2026 04:08:55 PM UTC
[09/21/26]seed@VM:~/.../internet-nano$
```

5. CHAIN OF CUSTODY

03:14:05 UTC - Acquisition - Labsetup.zip 2029834b (E01) -> Handler: Simon Friday Adeka

03:15:00 UTC - Extraction - Unzip verified (E02)

03:20:00 UTC - Build Failure - DNS 127.0.0.53 timeout documented (E03)

03:25:00 UTC - Build Success - Fix 8.8.8.8, Digest 9d65ed5a -> 170a7a0b8e75 (E04)

03:30:00 UTC - Network Create - 3 AS, 3 routers .254, Digest 7504e96a (E05)

03:32:00 UTC - Mapping - 7064=10.151.0.71 Attacker, b579=10.152.0.71 Victim (E06)

12:25:00 UTC - Code Analysis - Hash e8a77eb1 PID59 9090 (E07-E15)

16:08:55 UTC - Containment - docker ps empty, down verified (E16)

6. FINDINGS

1. Service identical on both hosts: PID 57 tail, PID 59 ./server LISTEN 9090 fd=3.
2. Worm in pre-weaponized state: NOP 500x, ret/offset 0x00, no badfile created.
3. Propagation via TCP 9090 netcat, not self-replicating in captured window.
4. Containment verified: 3 isolated bridges, no external egress, no /tmp artifacts.
5. Troubleshooting documented: DNS failure -> fix demonstrates real-world DFIR.

7. CONTAINMENT VERIFICATION

`docker-compose down` executed, `docker ps` returns empty, `date -u 16:08:55` timestamped, 3 bridge networks removed. Host filesystem `find badfile\*` empty confirms no weaponization.

8. CONCLUSION

Case closed as CONTAINED. 16 evidences meet NIST requirements: acquisition hash, troubleshooting, attribution via Docker ID/IP/PID correlation, code analysis, negative finding, propagation IoC, and verified cleanup. Ethical handling proven.

Analyst Signature: Simon Friday Adeka

Final Hash: e8a77eb1...afdb4ab

Repository: https://github.com/Simonadeka/CIP-B105-CS4-C11_26_DFIR_17300/evidence